

Compute the time complexity (Big O) for following problems.

1.

```
int a = 10, b = 20;
int sum = a + b;
for (int i = 0; i < n; i++)
    cout << i;
cout << sum;
```

Time Complexity: $O(n)$

2.

```
for (int i = 0; i < n; i++) {
    if (arr[i] == target)
        return i;
}
return -1;
```

Time Complexity: $O(n)$

3.

```
int count = 0;
for (int i = n; i > 1; i = i / 2) {
    count++;
}
```

Time Complexity: $O(\log n)$

4.

```
for (int i = 0; i < n; i++) {  
    sort(arr, arr + n);  
}
```

Time Complexity: $O(n^2)$

5.

```
for (int i = 0; i < n; i++) {  
    sort(arr, arr + n);  
}
```

Time Complexity: $O(n^2 \log n)$

6.

```
for (int i = 0; i < n; i++)  
    cout << i;
```

```
for (int i = 0; i < m; i++)  
    cout << i;
```

```
for (int i = 0; i < n; i++)  
    for (int j = 0; j < n; j++)  
        cout << i + j;
```

Time Complexity: $O(n^2 + m)$

